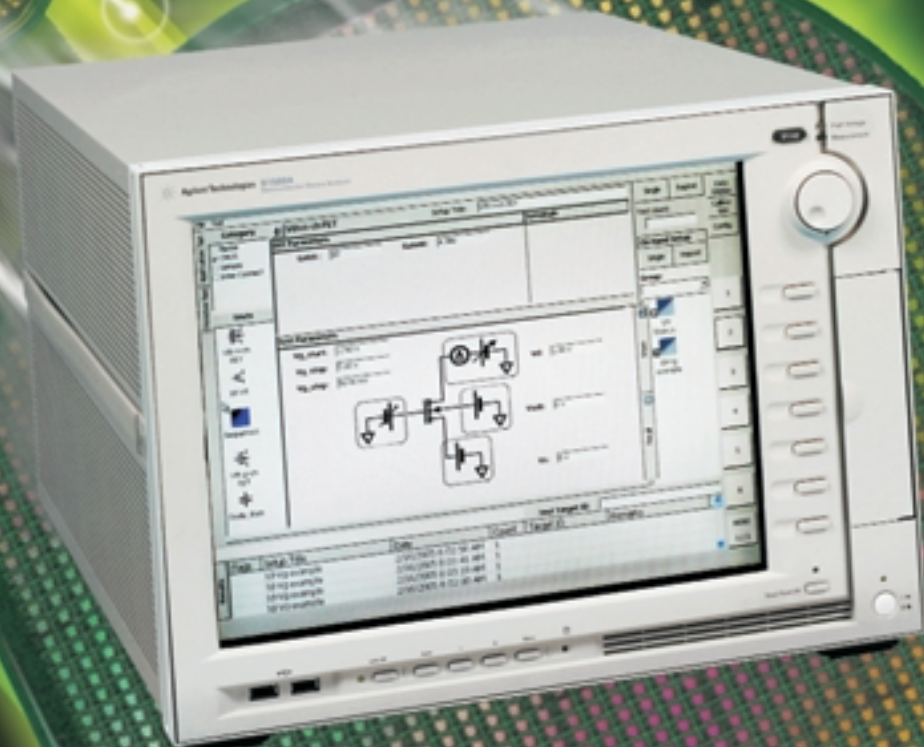


Making Every Engineer a Test Expert

Agilent's parameter analyser
teaches engineers to test



Top dogs

The industry's most
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Dream beam

Intel's laser
breakthrough

Made to measure

A new type of semiconductor parameter analyser will be needed to meet the needs of the industry at the 65nm and 45nm nodes, write **Alan Wadsworth** and **Tomoyuki Akiyama** of **Agilent Technologies**.

The history of the modern semiconductor parameter analyser began with the introduction of the HP 4145A in 1982. Before then, analogue curve tracers were the only available tools for semiconductor device analysis. The semiconductor parameter analyser introduced several revolutionary features. These included source/monitor units (SMUs), which can force and measure current and voltage, and the ability to capture and display data in a digital format. The analyser was so successful that the analogue curve tracer market virtually disappeared within a few years.

Since then, new versions of the semiconductor parameter analyser have been introduced and each generation has added new features, such as femtoamp current measurement resolution, reliability functions and quasi-static capacitance

versus voltage (QSCV) measurement. However, each new generation of parameter analyser has still followed the original concept of the 4145A, which required the user to have detailed knowledge of the functionality of the SMUs in order to create a simple test algorithm.

Although several Microsoft Windows PC-based solutions are available that support SMU set-up from a graphical user interface (GUI) environment, these current solutions still force the user to understand the functionality of the instrument hardware.

Challenges facing parametric test

Before the semiconductor industry crossed the 90nm lithography milestone, semiconductor process and device characterisation were much simpler.

The established semiconductor parameter analyser paradigm was sufficient to meet the industry's needs, mostly because users could rely on self-created measurement libraries to supplement those not directly supported by the analyser.

The next-generation 65nm processes and future 45nm semiconductor processes present many more parametric characterisation challenges, not only due to shrinking feature sizes but also factors such as high-k gate insulators, metal gate structures and low-k interconnect insulation materials.

Fundamental improvements in the semiconductor process characterisation workflow are needed to keep new process development on the International Technology Roadmap for Semiconductor (ITRS).

One of the most needed improvements is a more user-friendly software environment. Engineers and scientists cannot afford to invest time in learning the intricacies of the instrument hardware just to make a simple measurement. In

Figure 1. Picture of B1500A front showing innovative new task-oriented software environment



addition, many algorithms require a complicated test sequence in order to extract the desired data, requiring engineers and researchers to develop programming skills.

The second needed improvement is a better means to integrate current-voltage (IV) and capacitance-voltage (CV) measurements. Unlike IV, CV measurements are significantly affected by factors such as cable length, return path and device under test (DUT) impedance.

IV and CV measurement resources do not use the same types of connectors, so some sort of switching matrix hardware is necessary to connect to standard triaxial probes. Switching matrices however add complexity to the CV measurement environment.

In addition, new device structures such as silicon-on-insulator (SOI) transistors require support for pulsed IV measurements in the nanosecond range due to their increased susceptibility to harmful thermal effects during characterisation.

Next-generation solutions

The Agilent B1500A Semiconductor Device Analyser addresses these measurement challenges through improvements in software and hardware. The software provides an interface that allows the user to focus on the task at hand, without the need to learn the details of the instrument hardware. The B1500A utilises the Microsoft Windows XP Professional operating system, which makes it easy to integrate with PC-based work environments. The familiar and convenient Windows GUI reduces the learning curve and eliminates the need for instrument training. The B1500A hardware supports both IV and CV measurements within the same mainframe, eliminating the need to purchase a separate capacitance meter. Accurate and effortless switching between IV and CV measurements is achieved via an SMU CMU unify unit (SCUU), eliminating the need for an external switching matrix.

The B1500A software environment (Figure 1) uses a unique "top-down" approach to device characterisation. The user first selects one or more technology categories and then chooses an associated application test. Through a simple "fill in the blanks" process, a test algorithm can be quickly created. When this set-up process is complete, the user can begin taking parametric measurement data at the click of a button.

A library containing more than a hundred algorithms covering a wide variety of processes and device types is also included. It is also



possible to customise and save these algorithms into personal libraries and share them with various work groups. Users can also create their own unique algorithms completely from scratch.

The B1500A mainframe (Figure 2) supports a variety of SMU types (high-power, medium-power and high-resolution) and a multi-frequency capacitance measurement unit (MFCMU). Combining both IV and CV measurement into a single mainframe makes it easier to integrate the two functions.

Since capacitance measurement resources are calibrated to the outputs of the unit only, adding a cable to a capacitance measurement resource causes an error. Therefore, all measurements made through a cable must be "compensated" by applying various correction factors to the measurements.

The SCUU accepts a cabling fixture that connects to two of the SMUs and the CMU, and either MPSMUs or HRSMUs may be used. The cable assembly then connects to the actual SCUU, which is typically located close to the DUT.

The outputs of the SCUU consist of two triaxial connectors (Figure 3). This makes creation of an IV-CV measurement station simple and foolproof. In addition, the software takes care of all of the IV-CV switching, compensation and return path issues. The user just selects an IV or CV algorithm and pushes a button to begin making accurate measurements.

Measurement for the smaller geometry processes requires much more sensitive

Figure 2. Picture of B1500A rear showing various module types available (from top: 1xMFCMU, 2xMPSMU, 2xHRSMU, and 2xHPMSMU). The 4.2A ground unit (bottom) is standard

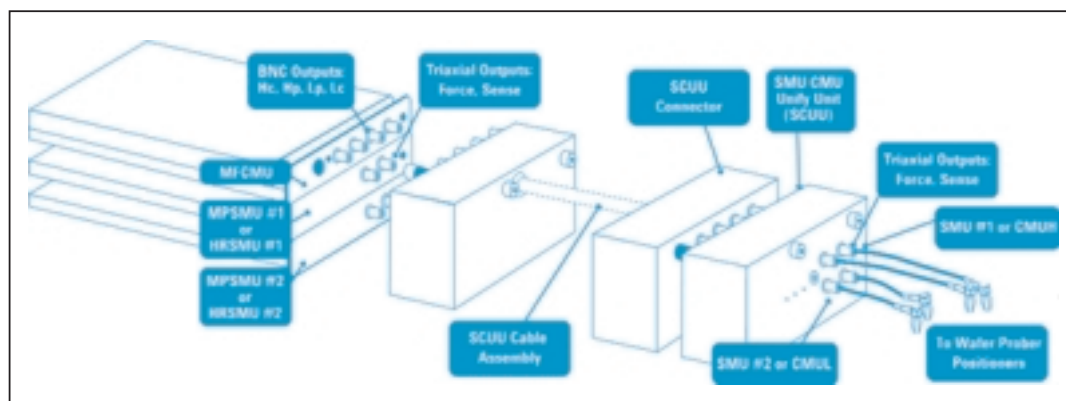


Figure 3. The SMU/CMU unify unit (SCUU) eliminates cabling confusion and prevents measurement errors

resolutions. For example, memory cells frequently need sub-1 femtoamp measurement resolution in order to precisely characterise parasitic leakage currents. To obtain this level of resolution, it is necessary to place some portion of the measurement hardware physically close to the DUT. The B1500A HRSMU accepts an optional atto sense and switch unit (ASU) for this purpose. The ASU increases the low-current measurement resolution of the HRSMU to 100 attoamps (0.1 femtoamp) without altering any other of the HRSMU's voltage and current measurement and forcing capabilities.

The B1500A ASU has two BNC inputs that are compatible with the outputs of a capacitance measurement unit, making it easy to switch between precision IV measurements and multi-frequency CV measurements without changing any cables. When used with the B1500A MFCMU, the ASU supports switching between CV measurements at up to 5MHz, and IV measurements with 0.1 femtoamp and 0.5 microvolt measurement resolution. As with the SCUU, the B1500A software takes care of all switching, compensation and return path issues.

The B1500A architecture makes it easy to add new solutions in the future to meet changing requirements. As the semiconductor industry crosses the 65nm and 45nm milestones and

requires extremely fast pulsed IV measurements, it is simple to add solutions that coordinate the B1500A measurement resources with external pulse generators such as the Agilent 81110A. In addition, it is also straightforward to add high-frequency CV solutions (>5MHz) using the Agilent 4294A and RF CV solutions (to 8.5GHz) with the Agilent E5071B.

Conclusion

The B1500A's software environment helps users to focus on device characterisation without having to learn the intricacies of the instrument hardware. This reduces start-up and training costs, shortening development cycle times. The B1500A integrates IV and CV analysis, to compensate all of the CV cable length and return path issues, and to make this integration invisible to the user. Engineers no longer have to struggle to design an IV and CV measurement solution, but instead can focus on device characterisation.

The flexibility of the B1500A software environment makes it easy to add new solutions in the future (such as high-frequency pulsed IV measurements) as changing process conditions require them. In short, the B1500A provides the characterisation capabilities to meet the needs of the semiconductor industry well into the next decade.



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Key to acronyms

IV = Current versus voltage
CV = Capacitance versus voltage
SMU = Source/monitor unit
HPSMU = High power source/monitor unit
MPSMU = Medium power source/monitor unit
HRSMU = High resolution source/monitor unit

CMU = Capacitance measurement unit
MFCMU = Multi-frequency capacitance measurement unit
SCUU = SMU/CMU unify unit
ASU = Atto-sense and switch unit
DUT = Device under test